

UNIT TEST — MATHEMATICS

MATHEMATICS BY D K SINGH

Topic: Matrices and Determinants

Class: XII (Board Level) — Total Questions: 40 — Max Marks: 120

General Instructions:

1. This paper consists of two sections: Section A (Multiple Choice Questions) and Section B (Integer Type Questions).
 2. Section A contains 30 Multiple Choice Questions (Q1 to Q30). Each question has four choices, out of which only one is correct.
 3. Section B contains 10 Integer Type Questions (Q31 to Q40). The answer to each question is a non-negative integer. Both sections each question carries 3 marks and no negative marking.
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SECTION A: MULTIPLE CHOICE QUESTIONS (30 Questions)

1. If $A = [a_{ij}]_{2 \times 2}$ where $a_{ij} = \frac{(i+2j)^2}{2}$, then the element a_{12} is:
(a) $\frac{9}{2}$
(b) $\frac{25}{2}$
(c) 8
(d) 18
2. For any square matrix A , the matrix $A - A^T$ is always a:
(a) Symmetric matrix
(b) Skew-symmetric matrix
(c) Diagonal matrix
(d) Identity matrix
3. If A is a square matrix of order 3 such that $|A| = 5$, then the value of $|\text{adj } A|$ is:
(a) 5
(b) 25
(c) 125
(d) $\frac{1}{5}$
4. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, then $A^2 - 5A$ is equal to:

- (a) $2I$
 - (b) $-2I$
 - (c) I
 - (d) O
5. The total number of possible matrices of order 3×3 with each entry 0 or 1 is:
- (a) 27
 - (b) 18
 - (c) 81
 - (d) 512
6. If A and B are symmetric matrices of the same order, then $AB - BA$ is a:
- (a) Skew-symmetric matrix
 - (b) Symmetric matrix
 - (c) Zero matrix
 - (d) Identity matrix
7. If $A = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$ and $A + A^T = I$, then the general value of α is:
- (a) $\frac{\pi}{6}$
 - (b) $\frac{\pi}{3}$
 - (c) π
 - (d) $\frac{3\pi}{2}$
8. If the area of a triangle with vertices $(2, -6)$, $(5, 4)$, and $(k, 4)$ is 35 sq. units, then k is:
- (a) 12
 - (b) -2
 - (c) 12 or -2
 - (d) -12 or -2
9. Let A be a non-singular square matrix of order 3×3 . Then $|3A|$ is equal to:
- (a) $3|A|$
 - (b) $9|A|$
 - (c) $27|A|$
 - (d) $81|A|$
10. If $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$, then A^{-1} is equal to:
- (a) $\frac{1}{19}A$
 - (b) $-\frac{1}{19}A$
 - (c) $\frac{1}{19}A^T$
 - (d) A

11. If A is a square matrix such that $A^2 = A$, then $(I + A)^3 - 7A$ is equal to:
- (a) A
 - (b) $I - A$
 - (c) I
 - (d) $3A$
12. If A and B are invertible matrices of the same order, then which of the following is correct?
- (a) $(AB)^{-1} = A^{-1}B^{-1}$
 - (b) $(AB)^{-1} = B^{-1}A^{-1}$
 - (c) $(AB)^T = A^T B^T$
 - (d) $(A + B)^{-1} = A^{-1} + B^{-1}$
13. If $\begin{bmatrix} x+y & 2 \\ 1 & x-y \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 1 & 7 \end{bmatrix}$, then the values of x and y are:
- (a) $x = 5, y = 2$
 - (b) $x = 5, y = -2$
 - (c) $x = -5, y = 2$
 - (d) $x = 2, y = 5$
14. If a matrix has 24 elements, what are the possible number of orders it can have?
- (a) 4
 - (b) 6
 - (c) 8
 - (d) 12
15. If A is a skew-symmetric matrix of order 3, then the value of $|A|$ is:
- (a) 0
 - (b) 1
 - (c) -1
 - (d) Any non-zero real number
16. If $\begin{vmatrix} x & 2 \\ 18 & x \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 18 & 6 \end{vmatrix}$, then x is equal to:
- (a) 6
 - (b) ± 6
 - (c) -6
 - (d) 0
17. If $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, then A^4 is equal to:
- (a) O
 - (b) I

- (c) A
 (d) $-I$
18. If A is a square matrix of order 3 and $|A| = k$, then $|\text{adj}(2A)|$ is:
- (a) $4k^2$
 (b) $16k^2$
 (c) $64k^2$
 (d) $8k^2$
19. For two matrices A and B , if $AB = O$, then:
- (a) $A = O$ or $B = O$
 (b) $A = O$ and $B = O$
 (c) It is not necessary that $A = O$ or $B = O$
 (d) All of the above
20. If $A = \begin{bmatrix} \lambda & 1 \\ -1 & -\lambda \end{bmatrix}$, then $A^2 = I$ for:
- (a) $\lambda = 0$ only
 (b) $\lambda = \pm 1$ only
 (c) no real value of λ
 (d) $\lambda = \pm\sqrt{2}$
21. The element in the second row and third column of the matrix inverse of $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}$ is found using its:
- (a) Cofactor C_{23}
 (b) Cofactor C_{32}
 (c) Minor M_{23}
 (d) Minor M_{32}
22. If $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, then A^5 is equal to:
- (a) $5A$
 (b) $10A$
 (c) $16A$
 (d) $32A$
23. If $A = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}$, $B = \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix}$ and $(A + B)^2 = A^2 + B^2$, then:
- (a) $a = 1, b = 4$
 (b) $a = 2, b = 3$

- (c) $a = 1, b = 3$
 (d) $a = 4, b = 1$
24. If the points $(x, -2)$, $(5, 2)$, and $(8, 8)$ are collinear, then the value of x using determinants is:
- (a) 3
 (b) 4
 (c) 5
 (d) 2
25. If A is an orthogonal matrix, then $|A|$ is equal to:
- (a) 0
 (b) ± 1
 (c) ± 2
 (d) None of these
26. If $A = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix}$, then the value of α for which $A^2 = B$ is:
- (a) 1
 (b) -1
 (c) 4
 (d) No real values
27. If A is a square matrix of order 3 and $A^T = A^{-1}$, then $|A|$ is:
- (a) ± 1
 (b) 0
 (c) 3
 (d) $\frac{1}{3}$
28. Let $f(x) = \begin{vmatrix} 1 & x & x+1 \\ 2x & x(x-1) & (x+1)x \\ 3x(x-1) & x(x-1)(x-2) & (x+1)x(x-1) \end{vmatrix}$, then $f(100)$ is equal to:
- (a) 100
 (b) 0
 (c) 5050
 (d) -100
29. If A is a row matrix as well as a column matrix, then the order of A is:
- (a) 1×1
 (b) $1 \times n$
 (c) $m \times 1$
 (d) None of these

30. If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, then $A^2 - 5A + 7I$ is equal to:

- (a) I
- (b) O
- (c) $2I$
- (d) $5I$

SECTION B: INTEGER TYPE QUESTIONS (10 Questions)

Note: The answer to each of the following questions is a single non-negative integer.

31. If $A = \begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix}$ where $i = \sqrt{-1}$, find the value of $|A^{40}|$.
32. If A is a square matrix of order 3 such that $|A| = 2$, find the value of $|\text{adj}(\text{adj } A)|$.
33. Find the non-negative integer value of x for which the matrix $\begin{bmatrix} x+1 & 2 \\ 3 & x-2 \end{bmatrix}$ is singular.
34. If $\begin{bmatrix} 2x & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -3 & 0 \end{bmatrix} \begin{bmatrix} x \\ 3 \end{bmatrix} = [0]$, find the positive integer value of x .
35. Let A be a 3×3 matrix such that $\det(A) = 4$. Find the value of $2 \cdot \det(2 \cdot A^{-1})$.
36. If the system of equations $x + y + z = 6$, $x + 2y + 3z = 10$, and $x + 2y + \lambda z = 10$ has infinitely many solutions, find the value of λ .
37. Let $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$. If $A^{10} = \begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$, find the value of k .
38. Find the maximum value of the determinant $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+x & 1 \\ 1 & 1 & 1+y \end{vmatrix}$ if $x, y \in \{1, 2, 3\}$.
39. If A is a square matrix of order 3 such that $A^2 = 2A$, and $|A| \neq 0$, find the value of $|A|$.
40. Find the value of k if the system of equations $2x + 3y = 5$ and $4x + ky = 10$ has an infinite number of solutions.

— END OF EXAMINATION —